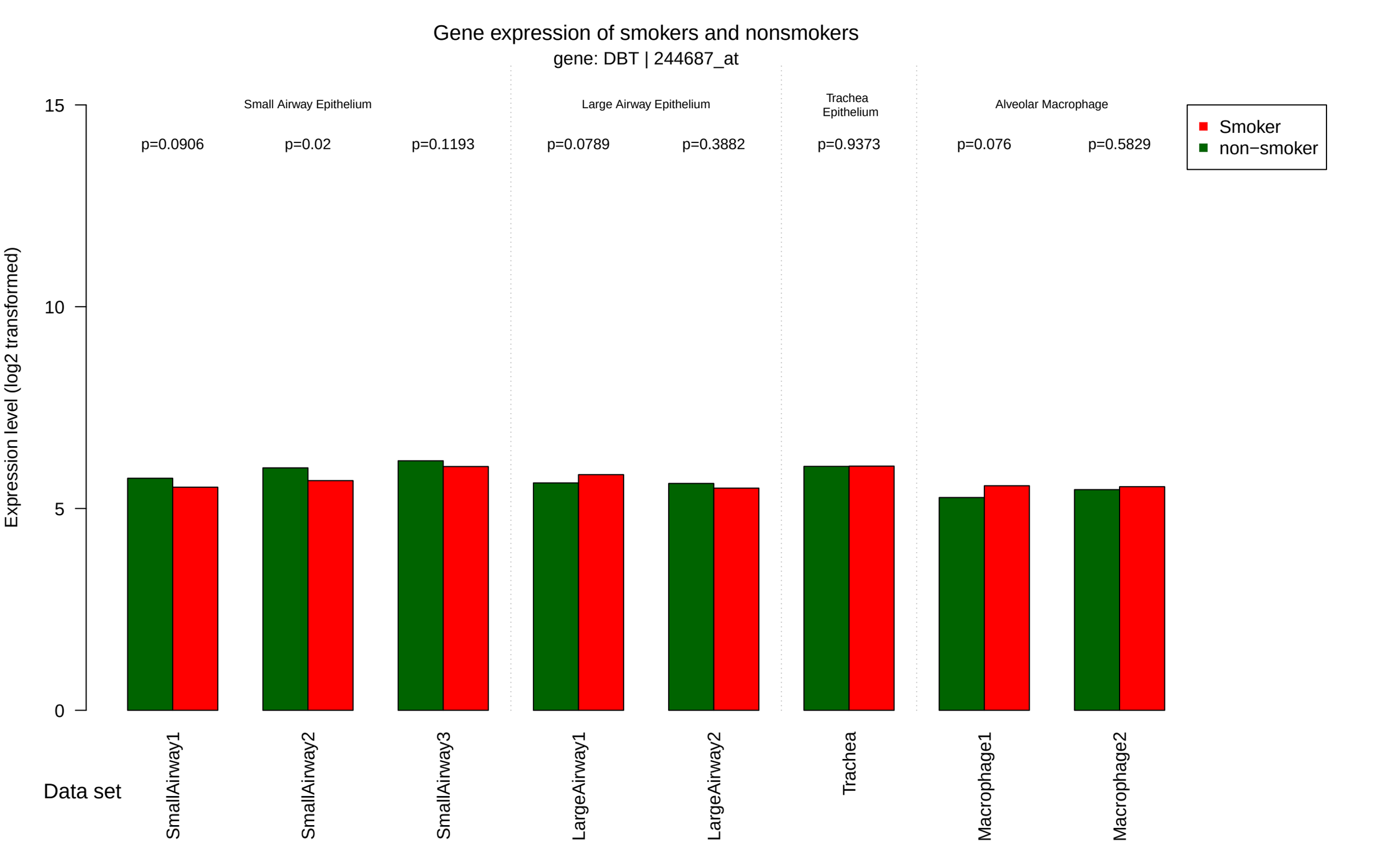
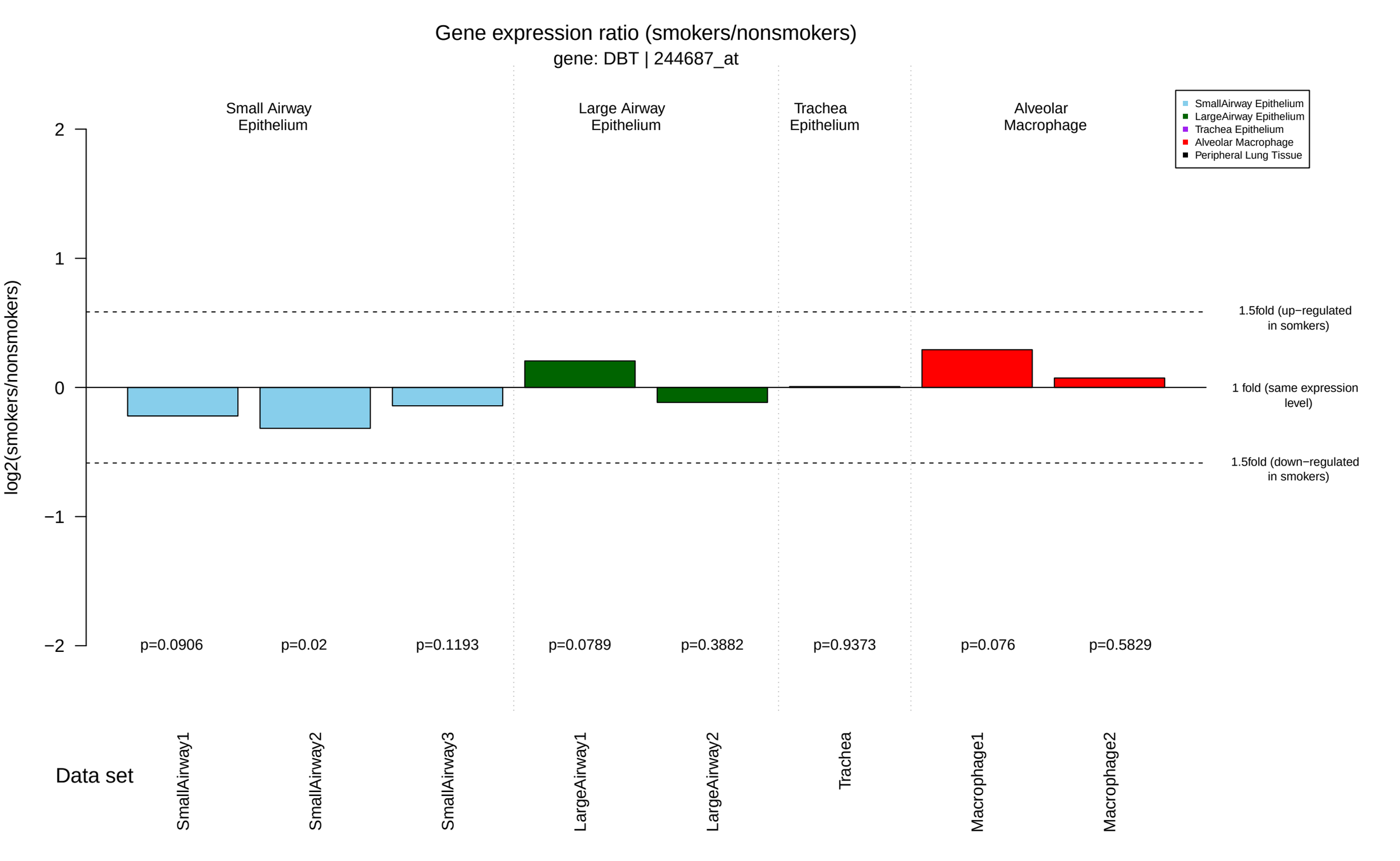
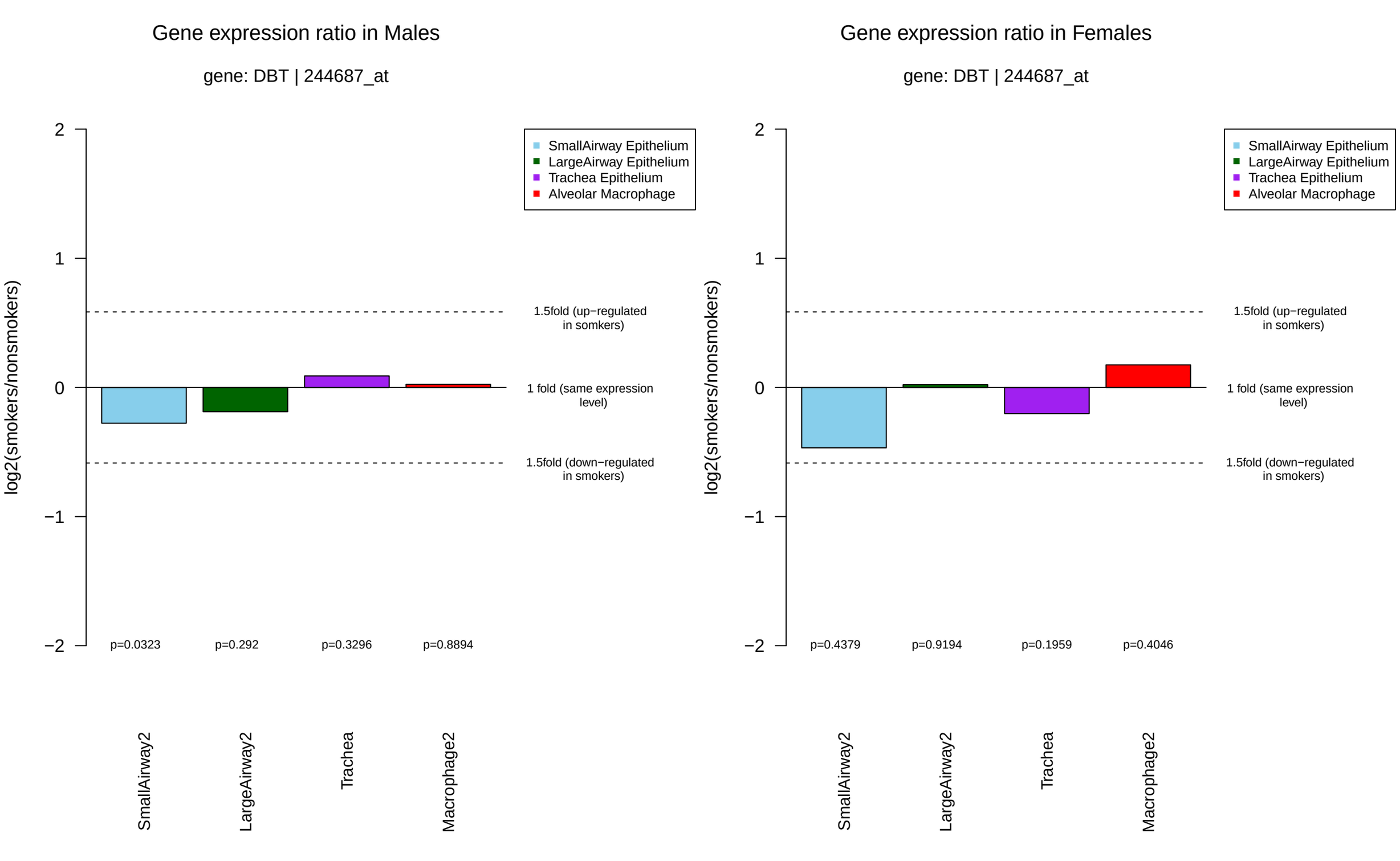
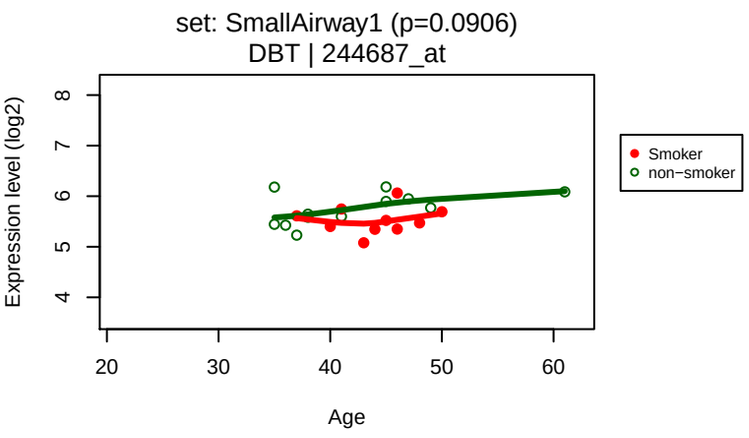
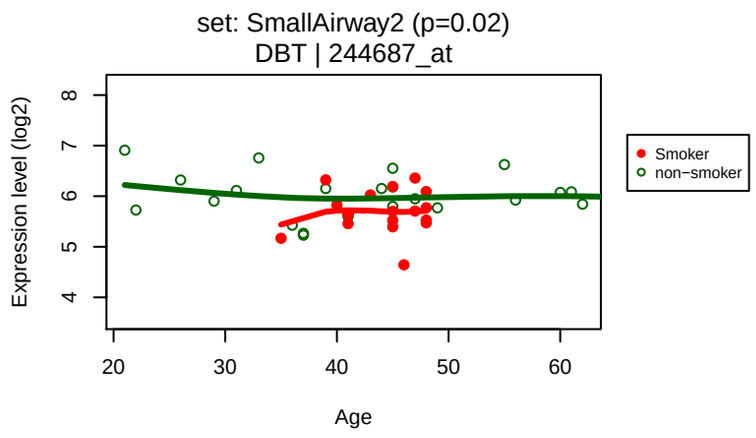
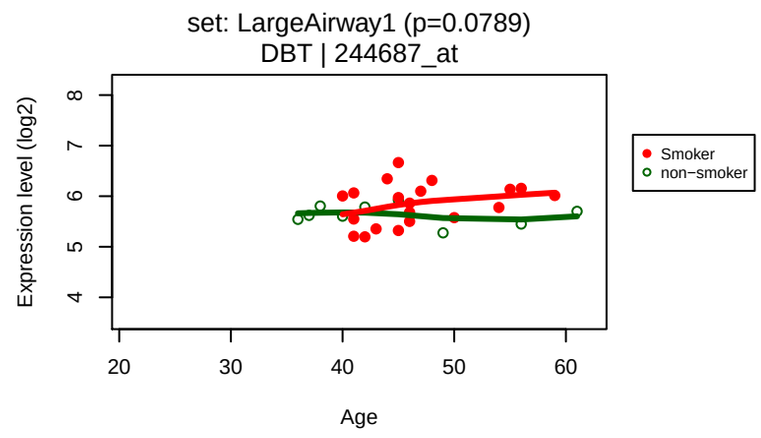

Gene expression of smokers and nonsmokers
gene: DBT | 244687_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0906
p=0.02
p=0.1193
p=0.0789
p=0.3882
p=0.9373
p=0.076
p=0.5829
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

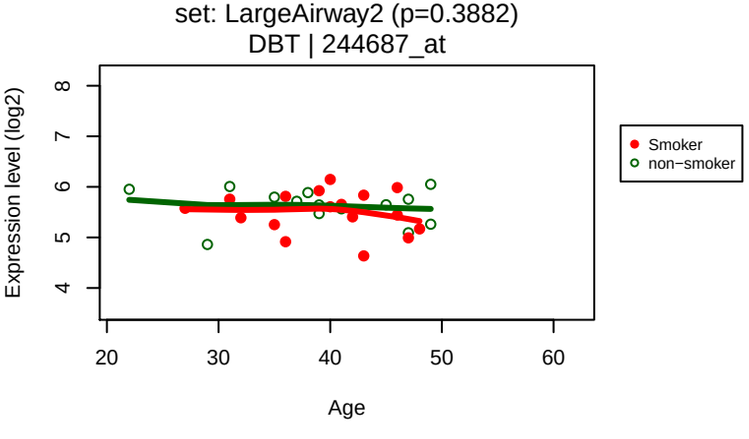

Gene expression ratio (smokers/nonsmokers)
gene: DBT | 244687_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0906
p=0.02
p=0.1193
p=0.0789
p=0.3882
p=0.9373
p=0.076
p=0.5829
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

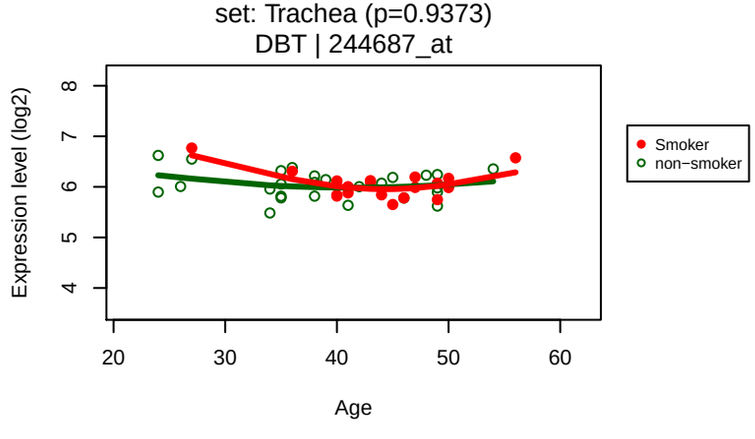

Gene expression ratio in Males
gene: DBT | 244687_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0323
p=0.292
p=0.3296
p=0.8894
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: DBT | 244687_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4379
p=0.9194
p=0.1959
p=0.4046
SmallAirway2
LargeAirway2
Trachea
Macrophage2

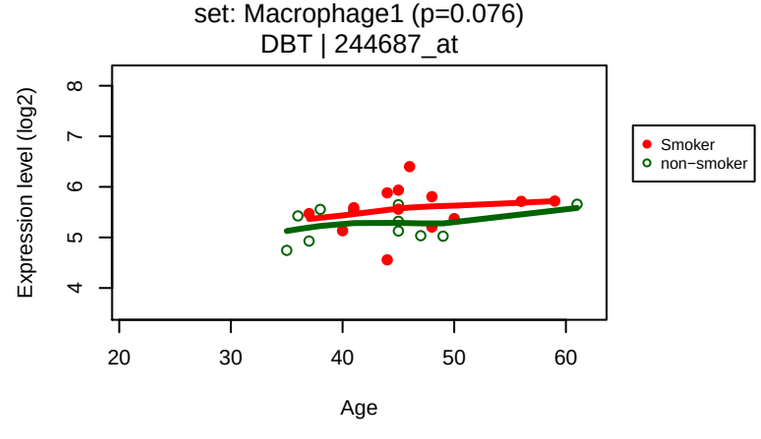

set: SmallAirway1 (p=0.0906)
DBT | 244687_at
Expression level (log2)
Age
Smoker
non−smoker

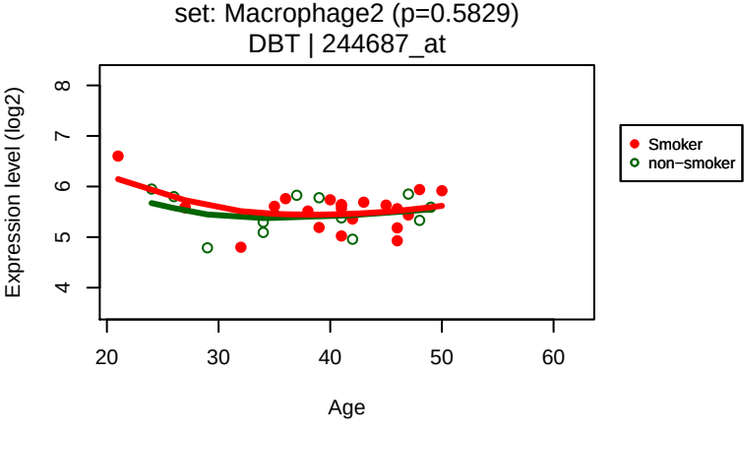

set: SmallAirway2 (p=0.02)
DBT | 244687_at
Expression level (log2)
Age
Smoker
non−smoker

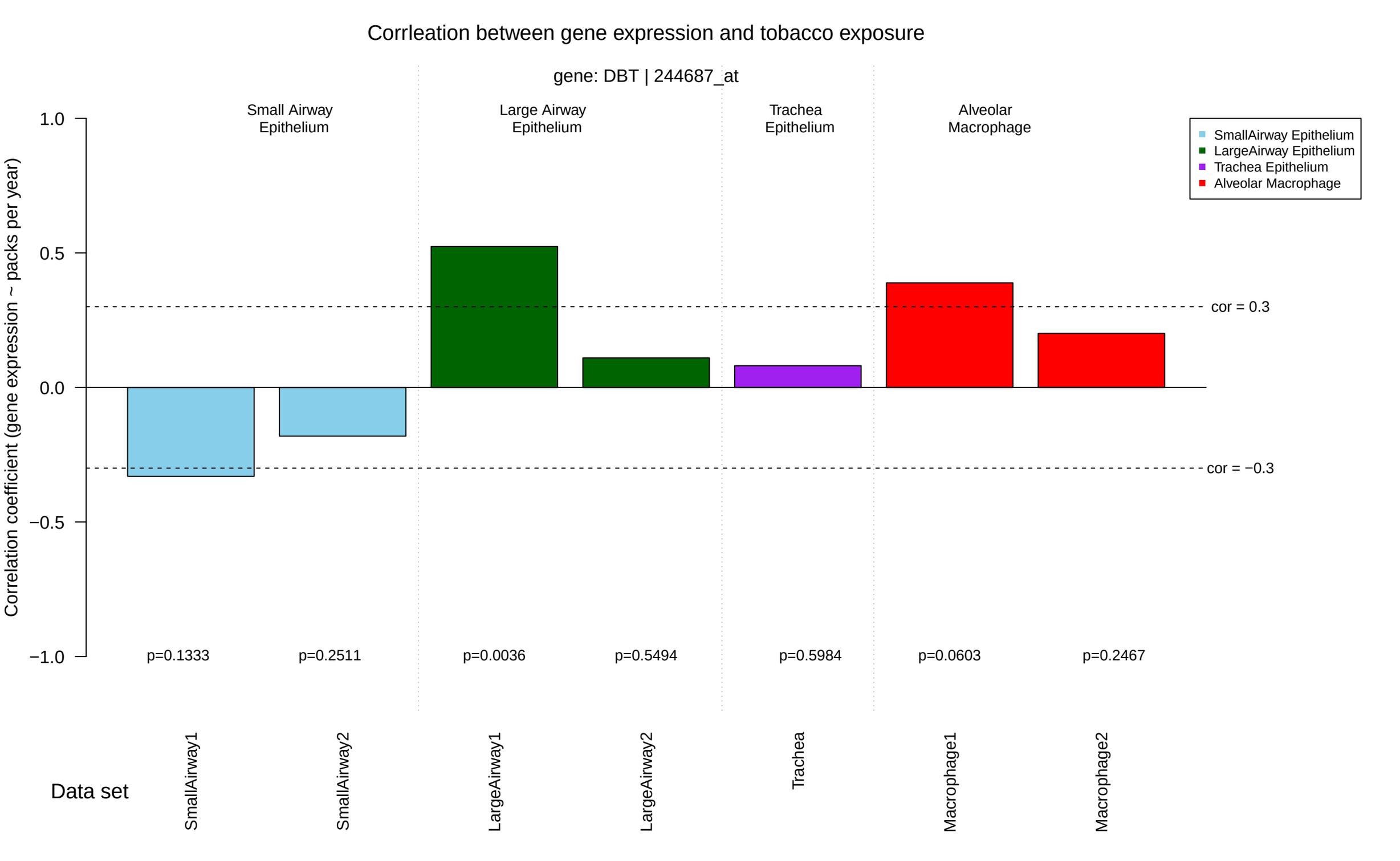

set: LargeAirway1 (p=0.0789)
DBT | 244687_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3882)
DBT | 244687_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9373)
DBT | 244687_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.076)
DBT | 244687_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5829)
DBT | 244687_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: DBT | 244687_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1333
p=0.2511
p=0.0036
p=0.5494
p=0.5984
p=0.0603
p=0.2467
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2